DAY 5

From Machine Learning to AI-Enabled Workflows

Evidence packages • AI-assisted pipelines • Guided LLM prompts • Human validation

Today's focus

Turn Day 3–4 model results into AI-ready evidence and reproducible, validated workflows.

Output

A grounded AI workflow concept and a prompt-chain approach for recreating an ML notebook.

Applied AI Design Workshop for Faculty
Fred Livingston (fjliving@ncsu.edu) 2026

Where we are in the workshop

Day 5 connects the foundational machine-learning workflow to AI-assisted implementation and interpretation.

DAY 3
Problem framing
Regression

DAY 4
Classification
Data preparation

DAY 5
AI assistance
Validated workflows

NEXT
Apply to teaching
research workflows

Progression

From building a defensible first-pass ML analysis → to using AI as a copilot for evidence-grounded communication, notebook generation, and workflow design.

Key shift: the LLM does not create new evidence; it helps organize, explain, and implement evidence that faculty can validate.

Day 3 recap: ML foundations and problem framing

Day 3 established the vocabulary and structure needed to create defensible ML problems.

Problem framing

Prediction objective
Inputs and outputs
Unit of observation
Target variable

Classic ML landscape

Regression
Classification
Clustering
Forecasting
Anomaly detection

Regression

Linear models
Polynomial trends
Regularization
Residuals and MAE/RMSE

Baseline thinking

Mean or majority baseline
Metric selection
Practical success criteria

Participant output

A framed ML problem with inputs, target, baseline, and metric.

What carries forward to Day 5?

- Every AI workflow still needs a clear unit of observation and target or discovery question.
- Baselines and metrics become the evidence that an AI explanation must respect.
- LLM-generated conclusions must be checked against the original ML framing.

Day 4 recap: from classification to data preparation

Day 4 expanded the workflow from supervised classification to data cleaning, feature preparation, and unsupervised discovery.

1 Classification

Predicted categorical outcomes using logistic regression, decision trees, probabilities, thresholds, and confusion matrices.

2 Baseline model activity

Built a first-pass model workflow with data inspection, baselines, simple models, metrics, and interpretation.

3 Data preparation

Addressed missing values, outliers, data types, encoding, scaling, leakage, and domain-informed features.

4 Unsupervised discovery

Used clustering, PCA, silhouette scores, DBSCAN, and cluster profiles to explore unlabeled data.

Day 4 generated the evidence packets that Day 5 will use:

Metrics
accuracy • recall • F1 • silhouette

Visuals
confusion matrices • PCA plots

Decisions
features • leakage • thresholds

Limits
data quality • validation • claims

Transition: model results are not yet an AI workflow

Day 5 asks how predictive evidence becomes context for AI-assisted analysis, communication, and notebook generation.

Data → Baseline → Model → Metrics → Evidence package → LLM workflow → Human review

Day 5 working principle

AI is useful when it is grounded in explicit evidence and constrained by human-validated assumptions, metrics, and limitations.

Do not ask only

"Can AI explain this?"

Ask instead

"What evidence can AI use, and what must it not claim?"

Day 5 schedule

BLOCK	FOCUS
Welcome and Day 5 framing	Connect Day 3–4 work to AI-enabled workflows
Session 1 — Evidence package synthesis	Problem, dataset, baseline, results, limitations, claim boundaries
Session 2 — AI across the full ML pipeline	Framing, audit, leakage, features, pipeline, evaluation, communication
Invited presentation — LLM and AI Workflows on NCSU's Hazel HPC	Scalable AI workflows and university computing resources
Break	
Session 3 — Guided LLM prompt chain	Seven-step prompt sequence with guardrails and human checkpoints
Session 4 — Worked example: recreate the ML notebook	Prompts → notebook → baseline/model results → model card → validation checks
Open discussion, questions	Faculty applications, technical questions, next steps

Day 5 learning objectives

By the end of Day 5, you should be able to use AI to support—not replace—a defensible ML workflow.

1 Synthesize evidence

Convert Day 3–4 results into an evidence package that includes metrics, visuals, limitations, and claim boundaries.

2 Map AI utility

Identify where AI can assist with problem framing, data auditing, leakage review, feature generation, code, evaluation, and explanation.

3 Prompt with guardrails

Use a guided prompt chain to generate notebook-ready markdown and code while preventing invented columns and unsupported claims.

4 Audit generated artifacts

Check LLM-generated notebooks for leakage, preprocessing errors, metric mismatch, hallucinations, and causal overclaims.

5 Communicate responsibly

Produce audience-specific explanations and model-card summaries grounded in evidence and explicit limitations.

6 Connect to implementation

Identify how AI workflows could support teaching, research, or lab operations while preserving human review.

Day 5 success criterion: participants can inspect, run, and defend AI-assisted ML artifacts.

Today’s learning activities

The activities are designed to move from evidence to implementation while keeping faculty in control of the analysis.

A Evidence package

Turn prior ML results into structured context that an AI system can use without inventing claims.

B AI pipeline map

Identify which stages are accelerated by AI and which require human validation or domain expertise.

C Prompt chain

Use seven controlled prompts to create notebook sections, code, metrics, interpretation, and a model card.

D Worked example

Follow a complete defect-classification example from prompt to notebook to validation checks.

Learning mode

Instructor demo

Show the evidence → prompt → notebook → validation workflow.

Faculty audit

Identify invented columns, leakage, metric mismatch, and causal overclaims.

Discussion

Translate the workflow to faculty research, teaching, and lab contexts.

Key phrase for the day: AI drafts the artifact; validate the assumptions, workflow, metrics, and claims.

Participant outputs for Day 5

The goal is not only to hear about AI workflows, but to leave with reusable structures and prompts.

AI evidence package

Problem, dataset, baseline, model result, limitations, and claim boundaries.

Prompt-chain recipe

A reusable seven-step sequence for generating a notebook scaffold.

AI audit checklist

Criteria for rejecting or revising AI-generated code and explanations.

Worked example notes

Defect-prediction workflow from dataset to logistic regression and model card.

Implementation idea

A faculty-specific teaching, research, or lab workflow where AI could assist.

Open questions

Technical, ethical, and organizational issues to address after the workshop.

These outputs are intentionally lightweight: they can become a course example, a research planning artifact, or a prototype workflow design.